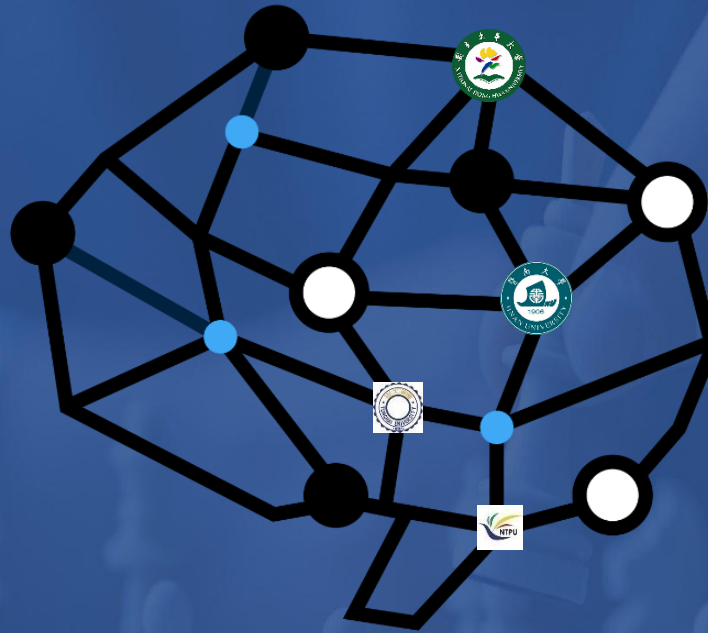




Chap. 10 First-Order Logic

課程：人工智慧導論
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First-order logic

- Propositional logic assumes the world contains **facts**.
- First-order logic (like natural language) assumes the world contains
 - **Objects**: people, houses, numbers, colors, baseball games, wars, ...
 - **Relations**: red, round, prime, brother of, bigger than, part of, comes between, ...
 - **Functions**: father of, best friend, one more than, plus, ...

Syntax of FOL: Basic elements

- Constants KingJohn, 2, NCNU,...
- Predicates Brother, $>$,...
- Functions Sqrt, LeftLegOf,...
- Variables $x, y, a, b, \dots$
- Connectives $\neg, \Rightarrow, \wedge, \vee, \Leftrightarrow$
- Equality $=$
- Quantifiers $\forall, \exists$

Atomic sentences

Atomic sentence = *predicate* ($term_1, \dots, term_n$)
or $term_1 = term_2$

Term = *function* ($term_1, \dots, term_n$)
or *constant* or *variable*

- E.g., *Brother*(KingJohn, RichardTheLionheart)
- $>$ (*Length*(LeftLegOf(Richard)), *Length*(LeftLegOf(KingJohn)))

Complex sentences

- Complex sentences are made from atomic sentences using connectives

$$\neg S, S_1 \wedge S_2, S_1 \vee S_2, S_1 \Rightarrow S_2, S_1 \Leftrightarrow S_2,$$

E.g. $Sibling(KingJohn, Richard) \Rightarrow$
 $Sibling(Richard, KingJohn)$

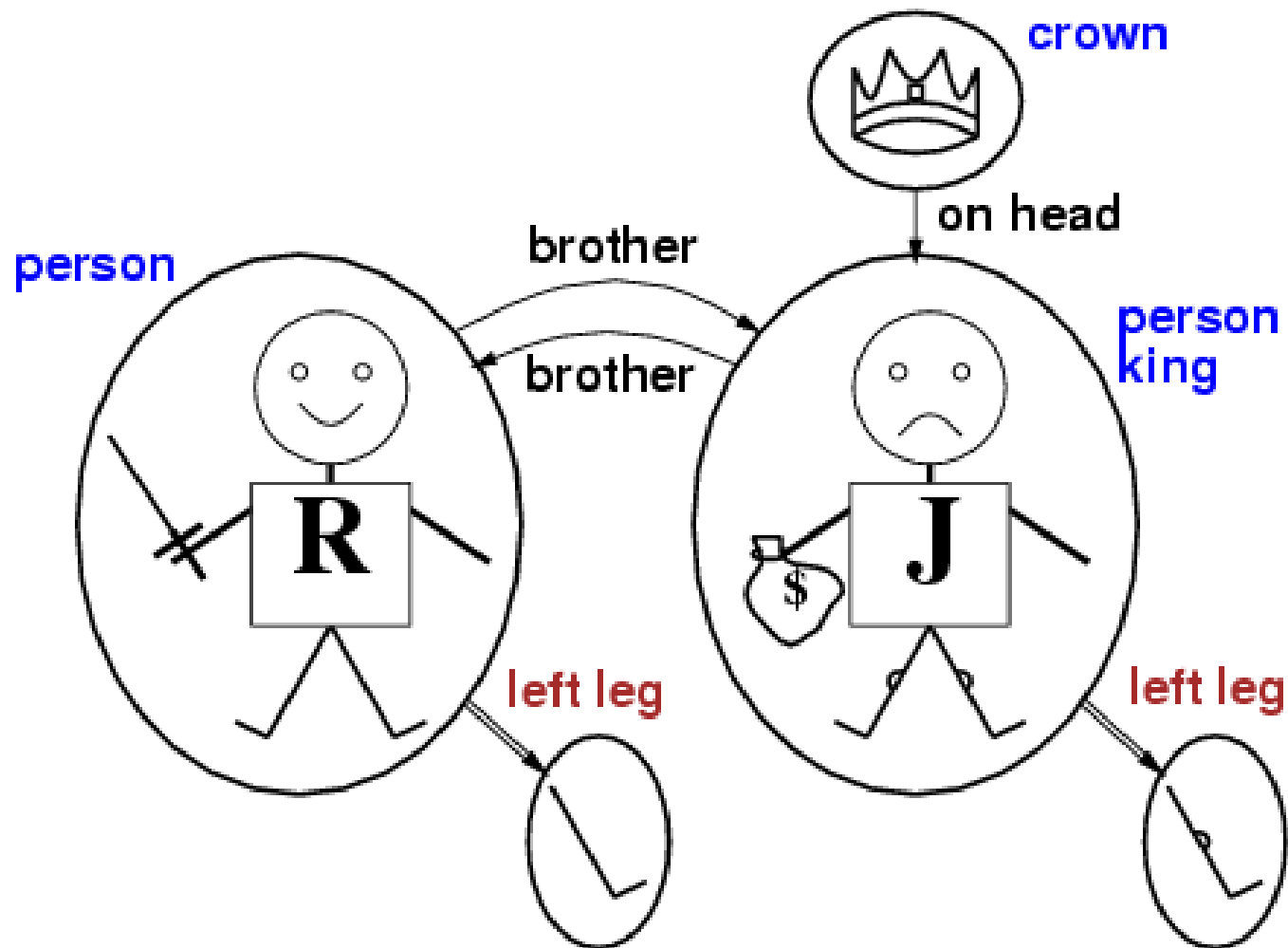
$$>(1,2) \vee \leq(1,2)$$

$$>(1,2) \wedge \neg >(1,2)$$

Truth in first-order logic

- Sentences are true with respect to a **model** and an **interpretation**
- Model contains **objects** (**domain elements**) and **relations** among them
- Interpretation specifies referents for
 - constant symbols** → **objects**
 - predicate symbols** → **relations**
 - function symbols** → **functional relations**
- An atomic sentence $predicate(term_1, \dots, term_n)$ is true iff the **objects** referred to by $term_1, \dots, term_n$ are in the **relation** referred to by $predicate$

Models for FOL: Example



Models for FOL

- We can enumerate the models for a given KB vocabulary:

For each number of domain elements n from 1 to ∞

For each k -ary predicate P_k in the vocabulary

For each possible k -ary relation on n objects

For each constant symbol C in the vocabulary

For each choice of referent for C from n objects ...

- Computing entailment by enumerating the models will not be easy !!

Quantifiers(量詞)

- Allows us to express properties of collections of objects instead of enumerating objects by name
- Universal: “for all” $\forall$
- Existential: “there exists” $\exists$

Universal quantification

$\forall \langle \text{variables} \rangle \langle \text{sentence} \rangle$

Everyone at NCNU is smart:

$$\forall x \text{ At}(x, \text{NCNU}) \Rightarrow \text{Smart}(x)$$

$\forall x P$ is true in a model m iff P is true with x being each possible object in the model

Roughly speaking, equivalent to the **conjunction** of instantiations of P

$$\begin{aligned} & \text{At}(\text{KingJohn}, \text{NCNU}) \Rightarrow \text{Smart}(\text{KingJohn}) \\ \wedge & \text{At}(\text{Richard}, \text{NCNU}) \Rightarrow \text{Smart}(\text{Richard}) \\ \wedge & \text{At}(\text{NCNU}, \text{NCNU}) \Rightarrow \text{Smart}(\text{NCNU}) \\ \wedge & \dots \end{aligned}$$

A common mistake to avoid

- Typically, $\Rightarrow$ is the main connective with $\forall$
 - **A universally quantifier is also equivalent to a set of implications over all objects**
- Common mistake: using $\wedge$ as the main connective with $\forall$:
 $\forall x \text{ At}(x, \text{NCNU}) \wedge \text{Smart}(x)$
means “Everyone is at NCNU and everyone is smart”

Existential quantification

$\exists \langle \text{variables} \rangle \langle \text{sentence} \rangle$

Someone at NCNU is smart:

$$\exists x \text{ At}(x, \text{NCNU}) \wedge \text{Smart}(x)$$

$\exists x P$ is true in a model m iff P is true with x being some possible object in the model

- Roughly speaking, equivalent to the **disjunction** of instantiations of P

$\text{At}(\text{KingJohn}, \text{NCNU}) \wedge \text{Smart}(\text{KingJohn})$

✓ **$\text{At}(\text{Richard}, \text{NCNU}) \wedge \text{Smart}(\text{Richard})$**

✓ **$\text{At}(\text{NCNU}, \text{NCNU}) \wedge \text{Smart}(\text{NCNU})$**

✓ ...

Another common mistake to avoid

- Typically, $\wedge$ is the main connective with $\exists$
- Common mistake: using $\Rightarrow$ as the main connective with $\exists$:

$$\exists x \text{ At}(x, \text{NCNU}) \Rightarrow \text{Smart}(x)$$

is true if there is anyone who is not at NCNU!

Properties of quantifiers

$\forall x \forall y$ is the same as $\forall y \forall x$

$\exists x \exists y$ is the same as $\exists y \exists x$

$\exists x \forall y$ is **not** the same as $\forall y \exists x$

$\exists x \forall y \text{ Loves}(x,y)$

□ **“There is a person who loves everyone in the world”**

$\forall y \exists x \text{ Loves}(x,y)$

□ **“Everyone in the world is loved by at least one person”**

■ **Quantifier duality**: each can be expressed using the other

$\forall x \text{ Likes}(x, \text{IceCream}) \quad \neg \exists x \neg \text{Likes}(x, \text{IceCream})$

$\exists x \text{ Likes}(x, \text{Broccoli}) \quad \neg \forall x \neg \text{Likes}(x, \text{Broccoli})$

Equality

- $term_1 = term_2$ is true under a given interpretation if and only if $term_1$ and $term_2$ refer to the same object
- E.g., definition of *Sibling* in terms of *Parent*:
$$\forall \mathbf{x}, \mathbf{y} \text{ **Sibling}(\mathbf{x}, \mathbf{y}) \Leftrightarrow [\neg(\mathbf{x} = \mathbf{y}) \wedge \exists \mathbf{m}, \mathbf{f} \neg (\mathbf{m} = \mathbf{f}) \wedge \text{Parent}(\mathbf{m}, \mathbf{x}) \wedge \text{Parent}(\mathbf{f}, \mathbf{x}) \wedge \text{Parent}(\mathbf{m}, \mathbf{y}) \wedge \text{Parent}(\mathbf{f}, \mathbf{y})]**}$$

Interacting with FOL KBs

- Suppose a wumpus-world agent is using an FOL KB and perceives a smell and a breeze (but no glitter) at $t=5$:

Tell(KB,Percept([Smell,Breeze,None],5))

Ask(KB, $\exists a$ BestAction($a,5$))

i.e., does the KB entail some best action at $t=5$?

- Answer: Yes, $\{a/Shoot\} \leftarrow$ substitution (binding list)
- Given a sentence S and a substitution α ,
- $S\alpha$ denotes the result of plugging α into S ; e.g.,
 - $S = \text{Smarter}(x,y)$**
 - $\alpha = \{x/Hillary,y/Bill\}$**
 - $S\alpha = \text{Smarter}(Hillary,Bill)$**
- **Ask(KB, S)** returns some/all α such that $KB \models S\alpha$.

Using FOL

The kinship domain:

- Brothers are siblings

$$\forall \mathbf{x}, \mathbf{y} \text{ *Brother*}(x, y) \Leftrightarrow \text{*Sibling*}(x, y)$$

- One's mother is one's female parent

$$\forall \mathbf{m}, \mathbf{c} \text{ *Mother*}(c) = m \Leftrightarrow (\text{*Female*}(m) \wedge \text{*Parent*}(m, c))$$

- “Sibling” is symmetric

$$\forall \mathbf{x}, \mathbf{y} \text{ *Sibling*}(x, y) \Leftrightarrow \text{*Sibling*}(y, x)$$

- A first cousin is a child of a parent's sibling

$$\forall \mathbf{x}, \mathbf{y} \text{ *FirstCousin*}(x, y) \Leftrightarrow \exists \mathbf{p}, \mathbf{ps} \text{ *Parent*}(p, x) \wedge \text{*Sibling*}(ps, p) \wedge \text{*Parent*}(ps, y)$$

Using FOL

The set domain:

- $\forall s \text{ Set}(s) \Leftrightarrow (s = \{\}) \vee (\exists x, s_2 \text{ Set}(s_2) \wedge s = \{x|s_2\})$
- $\neg \exists x, s \{x|s\} = \{\}$
- $\forall x, s \ x \in s \Leftrightarrow s = \{x|s\}$
- $\forall x, s \ x \in s \Leftrightarrow [\exists y, s_2 \} (s = \{y|s_2\} \wedge (x = y \vee x \in s_2))]$
- $\forall s_1, s_2 \ s_1 \subseteq s_2 \Leftrightarrow (\forall x \ x \in s_1 \Rightarrow x \in s_2)$
- $\forall s_1, s_2 \ (s_1 = s_2) \Leftrightarrow (s_1 \subseteq s_2 \wedge s_2 \subseteq s_1)$
- $\forall x, s_1, s_2 \ x \in (s_1 \cap s_2) \Leftrightarrow (x \in s_1 \wedge x \in s_2)$
- $\forall x, s_1, s_2 \ x \in (s_1 \cup s_2) \Leftrightarrow (x \in s_1 \vee x \in s_2)$

- $\{x|s\}$: the set resulting from adjoining element x to set s

FOL Version of Wumpus World

- Typical percept sentence:
Percept([Stench,Breeze,Glitter,None,None],5)
- Actions:
Turn(Right), Turn(Left), Forward, Shoot, Grab, Release, Climb
- To determine best action, construct query:
 $\forall a \text{ BestAction}(a,5)$
- ASK solves this and returns $\{a/\text{Grab}\}$

Knowledge base for the wumpus world

■ Perception

- $\forall \mathbf{b,g,t} \text{ Percept}([\mathbf{Smell,b,g}],\mathbf{t}) \Rightarrow \mathbf{Smelt(t)}$
- $\forall \mathbf{s,b,t} \text{ Percept}([\mathbf{s,b,Glitter}],\mathbf{t}) \Rightarrow \mathbf{Glitter(t)}$

■ Reflex

- $\forall \mathbf{t} \text{ Glitter(t)} \Rightarrow \mathbf{BestAction(Grab,t)}$

■ Reflex with internal state

- $\forall \mathbf{t} \text{ Glitter(t)} \wedge \neg \mathbf{Holding(Gold,t)} \Rightarrow \mathbf{BestAction(Grab,t)}$

Holding(Gold,t) can not be observed: keep track of change.

Deducing hidden properties

$\forall x,y,a,b \text{ Adjacent}([x,y],[a,b]) \Leftrightarrow$

$[a,b] \in \{[x+1,y], [x-1,y], [x,y+1], [x,y-1]\}$

Properties of locaton:

$\forall s,t \text{ At}(\text{Agent},s,t) \wedge \text{Smelt}(t) \Rightarrow \text{Smelly}(s)$

$\forall s,t \text{ At}(\text{Agent},s,t) \wedge \text{Breeze}(t) \Rightarrow \text{Breezy}(s)$

Squares are breezy near a pit:

□ **Diagnostic rule---infer cause from effect**

$\forall s \text{ Breezy}(s) \Rightarrow \exists r \text{ Adjacent}(r,s) \wedge \text{Pit}(r)$

□ **Causal rule---infer effect from cause (model based reasoning)**

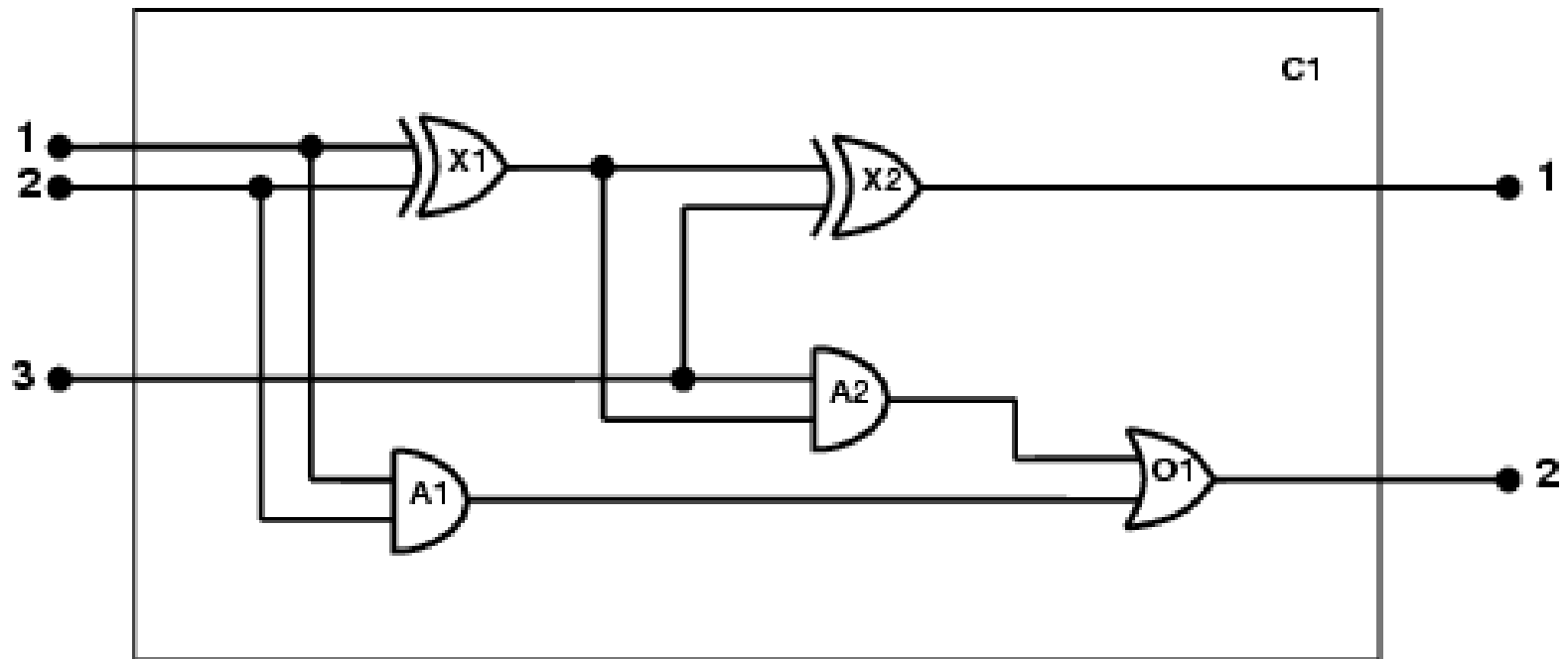
$\forall r \text{ Pit}(r) \Rightarrow [\forall s \text{ Adjacent}(r,s) \Rightarrow \text{Breezy}(s)]$

Knowledge engineering in FOL

1. Identify the task (what will the KB be used for)
2. Assemble the relevant knowledge
Knowledge acquisition.
3. Decide on a vocabulary of predicates, functions, and constants
Translate domain-level knowledge into logic-level names.
4. Encode general knowledge about the domain
define axioms
5. Encode a description of the specific problem instance
6. Pose queries to the inference procedure and get answers
7. Debug the knowledge base

The electronic circuits domain

One-bit full adder (and, or, xor)



The electronic circuits domain

- Identify the task
 - **Does the circuit actually add properly? (circuit verification)**
- Assemble the relevant knowledge
 - **Composed of wires and gates;**
 - **Types of gates (AND, OR, XOR, NOT)**
 - **Connections between terminals**
 - **Irrelevant: size, shape, color, cost of gates**
- Decide on a vocabulary
 - **Alternatives:**
 - Type(X_1) = XOR
 - Type(X_1 , XOR)
 - XOR(X_1)

The electronic circuits domain

4. Encode general knowledge of the domain

- $\forall \mathbf{t_1, t_2} \text{ Connected}(\mathbf{t_1, t_2}) \Rightarrow \mathbf{Signal(t_1) = Signal(t_2)}$
- $\forall \mathbf{t} \text{ Signal}(\mathbf{t}) = \mathbf{1} \vee \mathbf{Signal(t) = 0}$
- $\mathbf{1 \neq 0}$
- $\forall \mathbf{t_1, t_2} \text{ Connected}(\mathbf{t_1, t_2}) \Rightarrow \mathbf{Connected(t_2, t_1)}$
- $\forall \mathbf{g} \text{ Type}(\mathbf{g}) = \mathbf{OR} \Rightarrow \mathbf{Signal(Out(1,g)) = 1} \Leftrightarrow \exists \mathbf{n} \text{ Signal(In(n,g)) = 1}$
- $\forall \mathbf{g} \text{ Type}(\mathbf{g}) = \mathbf{AND} \Rightarrow \mathbf{Signal(Out(1,g)) = 0} \Leftrightarrow \exists \mathbf{n} \text{ Signal(In(n,g)) = 0}$
- $\forall \mathbf{g} \text{ Type}(\mathbf{g}) = \mathbf{XOR} \Rightarrow \mathbf{Signal(Out(1,g)) = 1} \Leftrightarrow \mathbf{Signal(In(1,g)) \neq Signal(In(2,g))}$
- $\forall \mathbf{g} \text{ Type}(\mathbf{g}) = \mathbf{NOT} \Rightarrow \mathbf{Signal(Out(1,g)) \neq Signal(In(1,g))}$

The electronic circuits domain

5. Encode the specific problem instance

Type(X_1) = XOR

Type(A_1) = AND

Type(O_1) = OR

Type(X_2) = XOR

Type(A_2) = AND

Connected(Out(1, X_1),In(1, X_2))

Connected(Out(1, X_1),In(2, A_2))

Connected(Out(1, A_2),In(1, O_1))

Connected(Out(1, A_1),In(2, O_1))

Connected(Out(1, X_2),Out(1, C_1))

Connected(Out(1, O_1),Out(2, C_1))

Connected(In(1, C_1),In(1, X_1))

Connected(In(1, C_1),In(1, A_1))

Connected(In(2, C_1),In(2, X_1))

Connected(In(2, C_1),In(2, A_1))

Connected(In(3, C_1),In(2, X_2))

Connected(In(3, C_1),In(1, A_2))

The electronic circuits domain

6. Pose queries to the inference procedure

What are the possible sets of values of all the terminals for the adder circuit?

$$\exists i_1, i_2, i_3, o_1, o_2 \text{ Signal(In(1, } C_1)) = i_1 \wedge \text{Signal(In(2, } C_1)) = i_2 \wedge \\ \text{Signal(In(3, } C_1)) = i_3 \wedge \text{Signal(Out(1, } C_1)) = o_1 \wedge \text{Signal(Out(2, } C_1)) = o_2$$

7. Debug the knowledge base

May have omitted assertions like $1 \neq 0$

A blue-tinted background image showing a hand moving a chess piece on a chessboard. The text "THE END." is centered in white, serif font, with a white horizontal line underneath it.

THE END.